

Symmetry and Groups

Week 1 · Lecture 1

Mehmet Özkan

Group Theory

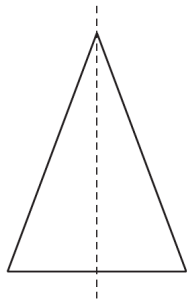
Reading for Lecture 1

- **Zee, I.1, pp. 37–43:** transformations, group axioms, cyclic examples, subgroups.
- **Bünemann, §1.2.1–1.2.2, pp. 3–5:** symmetries of bodies and a classical system.
- **Bünemann, §2.1, pp. 11–13:** axioms, group tables, and cyclic groups.

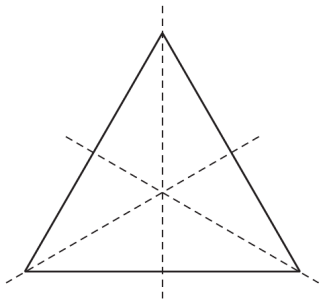
Work through the examples in these pages as we encounter each definition. The slide exercises give additional practice.

Group Theory

Zee's opening example: two triangles



(a)



(b)

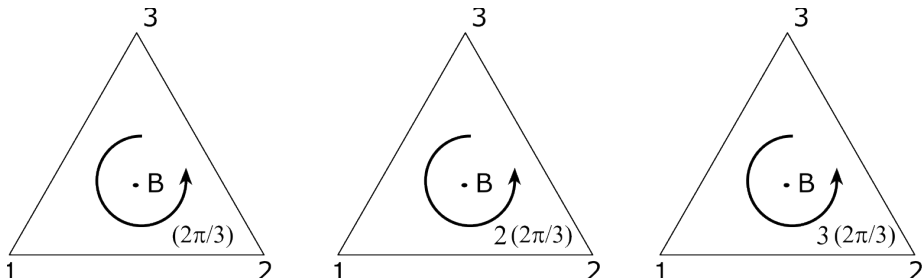
The dashed lines are mirror axes. The isosceles triangle has one; the equilateral triangle has three.

Which rotations also put the equilateral triangle back in place?

Figure 1

Group Theory

A first example: rotating a triangle

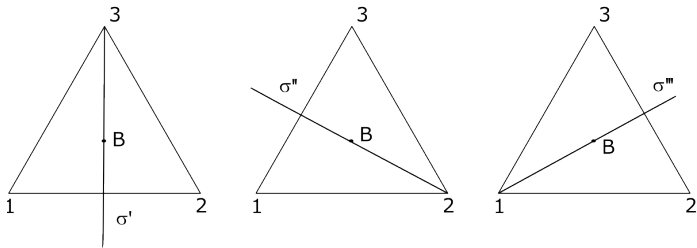


Turning the labeled triangle counterclockwise by 120° sends $1 \mapsto 2 \mapsto 3 \mapsto 1$. After three turns it returns to its starting position: $r^3 = e$. The rotations are $Z_3 = \{e, r, r^2\}$.

Image cropped from Grasso Luigi, *Equilateral triangle symmetries*, Wikimedia Commons, CC BY-SA 4.0.

Group Theory

A reflection of the triangle



Each mirror line passes through a vertex. The rightmost reflection s fixes 1, exchanges 2 with 3, and satisfies $s^2 = e$.

Image cropped from Grasso Luigi, *Equilateral triangle symmetries*, Wikimedia Commons, CC BY-SA 4.0.

Group Theory

What does a symmetry do?

- A symmetry is a transformation that leaves a chosen feature of a system unchanged.
- A square has four rotational symmetries. A generic rectangle has only two.
- The answer depends on the object *and* on what we ask to remain unchanged.

transform a system $\longrightarrow$ compare with the starting system.

Question: Which transformations preserve an equilateral triangle?

Group Theory

A notation guide to the two books

Zee writes I for the identity, Bünemann writes E , and we write e . They mean the same group element.

$$e = I = E \quad (\text{identity}), \quad |G| = \text{number of elements in finite } G.$$

We follow Zee's Z_n for a cyclic group of order n . Zee first gives its n th roots of unity under multiplication, then identifies them with the integers modulo n under addition.

$$Z_n = \{1, \omega, \dots, \omega^{n-1}\} \text{ (multiplicative realization),} \quad \{0, 1, \dots, n-1\} \text{ (additive realization).}$$

Bünemann calls this abstract group C_n . In additive calculations we use Z_n with the operation stated explicitly.

Group Theory

Composition is the operation

We multiply transformations by performing one after the other. Our convention throughout the course is

ab : apply b first, then apply a .

For example, $r^2r = r^3 = e$ and $rr^2 = e$. The identity e does nothing. Composition lets us ask how symmetries combine, rather than simply count them.

Group Theory

The multiplication table of Z_3

$\cdot$	e	r	r^2
e	e	r	r^2
r	r	r^2	e
r^2	r^2	e	r

The row supplies the left factor; the column supplies the right factor. Every answer remains in Z_3 .

Group Theory

A group: the definition

A **group** is a set G with a binary operation $(a, b) \mapsto ab$ such that

- **Closure:** $ab \in G$ whenever $a, b \in G$.
- **Associativity:** $(ab)c = a(bc)$.
- **Identity:** some $e \in G$ obeys $ea = ae = a$.
- **Inverses:** each $a \in G$ has $a^{-1} \in G$ with $aa^{-1} = a^{-1}a = e$.

Composing transformations is associative because both parenthesizations describe the same successive maps.

Group Theory

Checking the axioms for Z_3

- Products $r^a r^b = r^{a+b}$ reduce modulo 3: closure.
- The rotation $e = r^0$ is the identity.
- $r^{-1} = r^2$, $(r^2)^{-1} = r$.
- Rotations compose associatively because they are transformations.

$$r^a r^b = r^{a+b \bmod 3}.$$

The rotations form a group under composition.

Group Theory

Why associativity deserves its own axiom

Associativity concerns *parentheses*:

$$(ab)c = a(bc).$$

Commutativity concerns *order*:

$$ab = ba.$$

The group axioms require associativity. They do *not* require commutativity. We will find $ab \neq ba$ in the triangle's full symmetry group next class.

Group Theory

Cyclic groups

When repeated products of one element generate the entire group, write

$$\langle r \rangle = \{e, r, r^2, \dots\}.$$

If $r^n = e$ and no smaller positive power equals e , then

$$Z_n = \langle r \rangle = \{e, r, \dots, r^{n-1}\}, \quad |Z_n| = n.$$

The notation $|G|$ means the **order** of a finite group: the number of its elements.

Group Theory

The additive version of Z_n

We may label the elements $0, 1, \dots, n-1$ and add modulo n :

$$Z_n = \{0, 1, \dots, n-1\} \quad \text{under } + \pmod{n}.$$

The identity is 0. The inverse of a is $n-a$ modulo n .

$$Z_3 : \quad 1 + 1 = 2, \quad 1 + 2 = 0.$$

The correspondence $r^k \leftrightarrow k$ turns rotation composition into addition modulo n .

Group Theory

Zee's roots of unity and our rotations

Let $\omega = e^{2\pi i/3}$. Zee's Z_3 consists of $\{1, \omega, \omega^2\}$ under complex multiplication:

$$\omega^3 = 1, \quad \omega^a \omega^b = \omega^{a+b}.$$

The correspondence $r^k \mapsto \omega^k$ identifies the triangle's rotations with these roots of unity. We can also label the same three operations 0, 1, 2 and add modulo 3.

Textbook: Zee, I.1, pp. 41–42.

Group Theory

Two physical realizations, one multiplication law

- Rotating a triangle through 120° generates Z_3 .
- Shifting a three-site ring by one site also generates Z_3 .
- A group describes *how* transformations compose. The same group can act on different systems.

$$r^a r^b = r^{a+b \bmod 3} \quad \longleftrightarrow \quad a + b \bmod 3.$$

Group Theory

A physics example: parity

Consider a particle in one dimension:

$$H(x, p) = \frac{p^2}{2m} + V(x), \quad V(-x) = V(x).$$

The transformation $P : (x, p) \mapsto (-x, -p)$ leaves the Hamiltonian unchanged:

$$H(P(x, p)) = H(x, p), \quad P^2 = e.$$

Thus $\{e, P\} \cong Z_2$. This does not need quantum mechanics.

Group Theory

The empty action still counts

- The identity e belongs to every group. It maps every point to itself.
- An inverse reverses an operation: $g^{-1}g = e$.
- These properties make symmetry operations reversible.

For a rotation by 120° , the inverse is a rotation by 240° (or by -120°).

Group Theory

The order of an element

For $g \in G$, the **order of** g is the smallest positive integer m such that $g^m = e$, if one exists.

$$\text{ord}(g) = m \iff g^m = e \text{ for the first time at } m.$$

In Z_3 , $\text{ord}(e) = 1$ and $\text{ord}(r) = \text{ord}(r^2) = 3$. Do not confuse $\text{ord}(g)$ with $|G|$.

Group Theory

Worked example: Z_5

Here the operation is addition modulo 5; g^m in multiplicative notation becomes mg .

$$\langle 2 \rangle = \{0, 2, 4, 1, 3\} = Z_5.$$

Because $5 \cdot 2 = 0 \pmod{5}$ and no earlier positive multiple is zero, $\text{ord}(2) = 5$. Try the same calculation for $2 \in Z_6$.

Group Theory

Worked example: $2 \in Z_6$

Repeatedly add 2 modulo 6:

$$0, \quad 2, \quad 4, \quad 0.$$

Therefore $\langle 2 \rangle = \{0, 2, 4\}$ and $\text{ord}(2) = 3$. An element need not generate the whole group. We will call $\{0, 2, 4\}$ a **subgroup** in Week 2.

Group Theory

Which sets form groups?

Under the stated operation, decide whether each is a group.

1. Positive integers under addition.
2. All integers under addition.
3. Nonzero real numbers under multiplication.
4. All real numbers under multiplication.

For each failure, name the first axiom that fails.

Group Theory

Checking the four examples

1. Positive integers under $+$: no additive identity in the set.
2. $\mathbb{Z}$ under $+$: a group; inverse of a is $-a$.
3. $\mathbb{R} \setminus \{0\}$ under $\times$: a group; inverse is $1/a$.
4. $\mathbb{R}$ under $\times$: 0 has no multiplicative inverse.

The operation is part of the definition. The same underlying set can behave differently under different operations.

Group Theory

From rotations to reflections

The equilateral triangle also has reflections. A reflection changes the orientation of the vertex labels; a rotation does not.

$$\underbrace{e, r, r^2}_{\text{rotations}} \quad \underbrace{s, rs, r^2s}_{\text{reflections}}.$$

Next class we calculate how these six transformations compose and identify the resulting group with S_3 .

Reading: Zee, I.1, pp. 37–43; Bünemann, §1.2.1–1.2.2 and §2.1. Practice: Bünemann, Ch. 2, Ex. 1(a,b).

Dihedral and Permutation Groups

Week 1 · Lecture 2

Mehmet Özkan

Group Theory

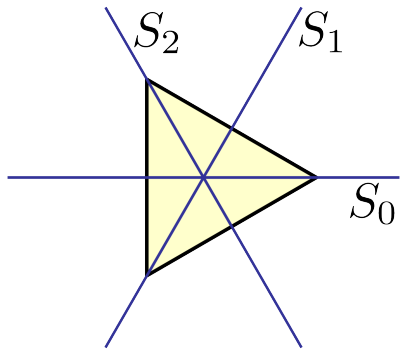
Reading for Lecture 2

- **Zee, I.1, pp. 46–50:** multiplication tables, the two groups of order four, and presentations.
- **Zee, I.2, pp. 55–57 and 60–61:** cycles, transpositions, and dihedral groups.
- **Bünemann, §2.2, pp. 13–17:** D_2 , C_4 , and D_3 as symmetries of solid bodies.

Bünemann's D_3 acts in three dimensions; the abstract multiplication rules agree with those of Zee's planar D_3 .

Group Theory

Three mirror axes, three turns



The equilateral triangle has three reflection axes. It also has three rotations that leave it in place:

$$e, \quad r, \quad r^2.$$

Counting both kinds of motion gives six symmetries. We now work out how they compose.

Image: Jim.belk, *Labeled Triangle Reflections*, Wikimedia Commons, public domain.

Group Theory

The full symmetry group of a triangle

Let r rotate the triangle by 120° , and let s reflect it across an axis through vertex 1.

$$r^3 = e, \quad s^2 = e, \quad srs = r^{-1}.$$

The full group is $D_3 = \{e, r, r^2, s, rs, r^2s\}$. There are three rotations and three reflections.

Group Theory

Why precisely six?

A symmetry sends vertex 1 to one of three vertices. Then it sends vertex 2 to one of the two remaining vertices. The image of vertex 3 is fixed.

$$3 \times 2 \times 1 = 6.$$

Every permutation of the vertices of an equilateral triangle is realized by one of its rotations or reflections.

Group Theory

The reflection reverses rotations

Reflect, rotate, and reflect again:

$$srs = r^{-1} = r^2.$$

Multiplying on the right by s gives $sr = r^{-1}s = r^2s$. Hence $sr \neq rs$: the full symmetry group is non-Abelian. The same relation holds for every regular n -gon, with $r^n = e$.

Group Theory

Reducing products to normal form

Every product can be written as r^k or $r^k s$, with k reduced modulo 3.

$$sr^k = r^{-k}s, \quad (r^a s)(r^b s) = r^{a-b}.$$

Example:

$$(rs)(r^2s) = r^{1-2} = r^2.$$

These rules suffice to construct the entire multiplication table without memorizing it.

Group Theory

The multiplication table of D_3

$\cdot$	e	r	r^2	s	rs	r^2s
e	e	r	r^2	s	rs	r^2s
r	r	r^2	e	rs	r^2s	s
r^2	r^2	e	r	r^2s	s	rs
s	s	r^2s	rs	e	r^2	r
rs	rs	s	r^2s	r	e	r^2
r^2s	r^2s	rs	s	r^2	r	e

Read, for instance, the entry in row s , column r : $sr = r^2s$.

Group Theory

The general dihedral group

The symmetries of a regular n -gon form

$$D_n = \langle r, s \mid r^n = e, s^2 = e, srs = r^{-1} \rangle.$$

Its $2n$ elements are

$$e, r, \dots, r^{n-1}, \quad s, rs, \dots, r^{n-1}s.$$

This notation calls the group of the square D_4 , of order 8. Some books call the same group D_8 ; we always use $|D_n| = 2n$.

Group Theory

Two geometric realizations of D_3

Zee, I.2, pp. 60–61: rotations and reflections in the plane preserve an equilateral triangle.

Bünemann, §2.2.3, pp. 15–17: proper rotations in space preserve a triangular prism. The three half-turns play the algebraic role of the planar reflections.

Both realize the same six-element group. The geometric transformations differ.

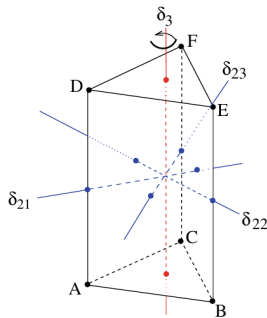


Image: Bünemann, Fig. 2.3, p. 16.

Group Theory

Why D_4 is larger than Z_4

The square's rotation subgroup is $Z_4 = \{e, r, r^2, r^3\}$, with r a quarter-turn. The full symmetry group also includes four reflections.

$$|Z_4| = 4, \quad |D_4| = 8.$$

In Z_4 every pair of elements commutes. In D_4 , $sr = r^{-1}s \neq rs$.

Group Theory

The two groups of order four

Bünemann's $C_4 = \langle r \rangle$ has an element r of order four:

$$C_4 = \{e, r, r^2, r^3\}, \quad r^4 = e.$$

His D_2 consists of the identity and three half-turns of a cuboid with unequal edges. Each nonidentity element squares to e .

$$D_2 = \{e, a, b, ab\}, \quad a^2 = b^2 = e, \quad ab = ba.$$

The groups have the same size but cannot be isomorphic: D_2 has no element of order four.

Textbook: Zee, I.1, pp. 47–50; Bünemann, §2.2.1–2.2.2, pp. 14–15.

Group Theory

A matrix check of $sr s = r^{-1}$

In the plane, choose s as reflection across the x axis and r as rotation by $\theta = 2\pi/n$:

$$S = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad R = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

Direct multiplication gives $SRS = R^{-1} = R(-\theta)$.

Group Theory

Permutations of three objects

A permutation is a bijection from a set to itself. S_3 contains all permutations of $\{1, 2, 3\}$.

$$|S_3| = 3! = 6.$$

We compose permutations from **right to left**, exactly as we compose the triangle symmetries.

Group Theory

Cycle notation

The cycle (123) sends $1 \mapsto 2$, $2 \mapsto 3$, $3 \mapsto 1$. The transposition (23) exchanges 2 and 3 while fixing 1.

$$(123)^3 = e, \quad (23)^2 = e.$$

In S_3 the complete list is e , the two 3-cycles (123) , (132) , and the three transpositions (12) , (13) , (23) .

Group Theory

Transpositions and parity in S_3

A transposition exchanges two labels. Zee shows that permutations can be built from transpositions.

$$(123) = (12)(23), \quad (132) = (23)(12).$$

The identity and the two 3-cycles use an even number of transpositions; the three exchanges use an odd number. **In Week 2 we will turn even/odd parity into a homomorphism.**

Textbook: Zee, I.2, pp. 56–57.

Group Theory

The triangle identifies D_3 with S_3

Label the triangle counterclockwise, starting at a vertex on the reflection axis. Then

$$r = (123), \quad s = (23), \quad r^2 = (132),$$

$$rs = (12), \quad r^2s = (13).$$

Every triangle symmetry produces a different permutation, and every permutation occurs. Composition agrees in both descriptions.

Group Theory

Check one product by acting on labels

Set $r = (123)$ and $s = (23)$. To compute rs , apply s first:

$$1 \xrightarrow{s} 1 \xrightarrow{r} 2, \quad 2 \xrightarrow{s} 3 \xrightarrow{r} 1, \quad 3 \xrightarrow{s} 2 \xrightarrow{r} 3.$$

Thus $rs = (12)$, while $sr = (13)$. We have displayed a concrete failure of commutativity.

Group Theory

Order in S_3

$$\text{ord}(e) = 1, \quad \text{ord}((123)) = 3, \quad \text{ord}((12)) = 2.$$

Every element's order tells us how many repetitions return it to the identity. A 3-cycle is a rotation; a transposition acts as a reflection on the triangle.

Group Theory

An exercise in composition

Use right-to-left composition. Calculate

$$(12)(23), \quad (23)(12), \quad (123)(23).$$

For the first product, write down the images of 1, 2, and 3 separately. Do the first two answers agree?

Group Theory

Composition: answers

$$(12)(23) = (123), \quad (23)(12) = (132),$$

$$(123)(23) = (12).$$

The first two are distinct. A multiplication convention must always be stated before using cycle notation.

Group Theory

Why these examples matter in physics

- Z_n describes rotations of a regular ring of n sites.
- Planar D_n adds reflections of that ring.
- S_n describes permutations of n labels. Exchanging identical constituents is a symmetry when the dynamics respects the exchange.

The group fixes the algebraic possibilities before we introduce states or representations.

Group Theory

Two questions for next week

Within D_3 , the rotations $\{e, r, r^2\}$ form a group of their own. So does $\{e, s\}$.

- Which subsets of a group are themselves groups?
- Why do their sizes divide the size of the full group?

These lead to subgroups and Lagrange's theorem.

Reading: Zee, I.1, pp. 46–50 and I.2, pp. 55–57, 60–61; Bünemann, §2.2. Challenge: Bünemann, Ch. 2, Ex. 8.